

# Chapter 5 Production Workflow & Evidence Plan

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## Intelli-Trading Fours Dissertation (Evaluation Chapter)

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### 0. Purpose of This Document

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This document defines a **strict, executable workflow** for producing Chapter 5 (Evaluation).

It is designed to:

- Eliminate vague writing and “hand-waving”
  - Force **evidence-first reasoning**
  - Ensure **complete coverage of claims**
  - Make the chapter reproducible by any agent or collaborator
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### 1. Core Evaluation Philosophy

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#### 1.1 What Chapter 5 IS

A structured, evidence-driven technical investigation of whether the system behaves as designed.

The system is evaluated as:

- an **interaction architecture**
  - a **closed-loop musical system**
  - a **real-time responsive pipeline**
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#### 1.2 What Chapter 5 is NOT

- Not a narrative reflection
  - Not a feature description
  - Not a list of tests
  - Not a claim of musical “quality” without evidence
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#### 1.3 Evaluation Lens (CRITICAL)

All evaluation must answer:

Does the system demonstrate **meaningful, interpretable, responsive interaction**, rather than just output generation?

This aligns with co-creative system research, where:

- systems must **relate to user input to avoid appearing random**

- interaction quality depends on **understandability and responsiveness**, not just output
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## 2. Chapter 5 Global Workflow

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DO NOT SKIP OR REORDER THESE PHASES

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### Phase 0 — Claim Ledger (MANDATORY START)

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#### Goal

Define EXACTLY what the chapter proves.

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#### Output Format

##### Strongly Evidenced Claims

- Fully supported by existing system + data

##### Partially Evidenced Claims

- Some evidence exists, but incomplete

##### Explicitly Deferred Claims

- Not evaluated (must be clearly stated)
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### Example Categories (pre-filled baseline)

#### Strong

- Closed-loop interaction exists
- Turn-taking architecture works
- Planner produces interpretable outputs
- Structural response differences exist

#### Partial

- Real-time responsiveness (needs timing evidence)
- Plan → output alignment
- Musical plausibility

#### Deferred

- User studies
- Long-term adaptation

- Full expressive generation

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## GATE

No writing begins until ALL claims are categorised.

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## Phase 1 — Evidence Register

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### Goal

Track every usable asset.

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### Required Table

| ID | Asset | Type | Claim | Status | Section | Notes |
|----|-------|------|-------|--------|---------|-------|
|----|-------|------|-------|--------|---------|-------|

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### Asset Types

#### Tests

- unit tests (analysis, planner, skeleton)
- integration tests (loop)

#### Batch Data

- planner CSV
- skeleton outputs
- tuning logs

#### Debug Tools

- PatternTurn visualiser
- Planner debug UI
- Skeleton debug UI
- TurnLoop state UI

#### Historical Evidence

- pre/post tuning comparisons

#### Report Evidence

- interim report claims (scope + limitations)
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## GATE

If it’s not in the register → it cannot be used.

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## Phase 2 — Section Planning Grid

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Each section must be planned BEFORE writing.

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### Required Template

| Section | Question | Claims | Evidence IDs | Figures | Tables | Missing |
|---------|----------|--------|--------------|---------|--------|---------|
|---------|----------|--------|--------------|---------|--------|---------|

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### GATE

No section is drafted without:

- at least 1 evidence source
  - at least 1 figure/table (unless justified)
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## Phase 3 — Visual-First Design

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### Principle

Figures and tables define the chapter. Text explains them.

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### Required Artefacts

#### Tables

1. Evaluation Questions Mapping
2. Subsystem Summary
3. Planner Case Table
4. Skeleton Metrics Summary
5. Claim Evidence Table

#### Figures

1. Planner distribution
  2. Planner margin/confidence
  3. Planner traceability example
  4. Skeleton comparisons
  5. Runtime debug screenshot
  6. PatternTurn visualisation
  7. Planner debug panel
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## GATE

Every section must reference at least one artefact.

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## Phase 4 — Addition Decisions

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Identify missing evidence BEFORE writing.

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### Required Decisions

#### A. Plan → Output Alignment

Compare:

- input
- plan
- skeleton
- final output

#### B. Timing Evidence

Measure:

- capture → plan → playback latency

#### C. Case Studies

Select 3–5 representative examples

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For each:

- DO
  - DON'T
  - DEFER
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## GATE

No “implied evaluation” allowed — decisions must be explicit.

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## Phase 5 — Asset Pack Assembly

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### Goal

Create a single source of truth.

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## Must Contain

- selected datasets
  - selected figures
  - selected cases
  - selected screenshots
  - selected limitations evidence
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## GATE

No drafting from memory — only from Asset Pack.

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## Phase 6 — Drafting Order (STRICT)

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1. 5.1 Framework
  2. 5.2 Subsystems
  3. 5.3 Planner (deep)
  4. 5.4 Skeleton (deep)
  5. 5.5 Runtime
  6. 5.6 Limitations
  7. 5.7 Synthesis
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## Rule

Core sections (5.3, 5.4) must be strongest.

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## Phase 7 — Red-Team Pass

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## Checklist

For every paragraph:

- Is the claim classified (strong/partial/deferred)?
  - Is evidence explicitly referenced?
  - Is this describing design (should be Chapter 3)?
  - Is this overclaiming musical quality?
  - Should this be a figure instead?
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## GATE

Unjustified claims are removed or downgraded.

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## 3. Chapter Structure (Final)

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### 5.1 Evaluation Framework

Define scope, claims, and evaluation approach.

### 5.2 Subsystem Correctness

Verify pipeline components function correctly.

### 5.3 Planner Behaviour

Evaluate interpretability and decision logic.

### 5.4 Structural Response Distinctiveness

Evaluate generated structures (not audio quality).

### 5.5 Closed-Loop Execution

Evaluate real-time loop and observability.

### 5.6 Limitations

Explicit gaps between architecture and output.

### 5.7 Synthesis

What is proven vs not proven.

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## 4. Critical Writing Rules

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### Rule 1 — Evidence First

No claim without explicit evidence.

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### Rule 2 — No README Writing

Descriptions must support arguments.

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### Rule 3 — No Output Overclaiming

System is NOT a finished musical agent.

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### Rule 4 — Always Separate Layers

- Plan ≠ Skeleton ≠ Output
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## Rule 5 — Interaction > Generation

Focus on responsiveness and structure.

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## 5. Theoretical Anchoring (for justification)

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Use literature to justify evaluation framing:

- Co-creative systems must maintain **relation to user input**
  - Call-and-response is fundamental to musical interaction structure
  - Interactive systems require **real-time adaptation and coordination**
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## 6. Final Execution Summary

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### Workflow

1. Build Claim Ledger
  2. Build Evidence Register
  3. Plan Sections
  4. Design Figures/Tables
  5. Decide Additions
  6. Assemble Asset Pack
  7. Draft in Order
  8. Red-team pass
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### Core Principle

This chapter does not “argue quality” — it **demonstrates system behaviour under controlled evidence.**

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END OF DOCUMENT